

CSA10 - CO4 Test Case Design Assignment

"Injury Risk Prediction System for Athletes Using Performance Data"

CO4 ASSESSMENT TOOL 1 - TEST CASE DESIGN ASSIGNMENT

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Submitted by

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DECLARATION

I, Dilli babu N, of the Department of Computer Science and Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the assignment work entitled “Injury Risk Prediction System for Athletes Using Performance Data - Test Case Design” is the result of my own bonafide effort. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

Place: Chennai

Date: 18/08/2026

Signature of the Student

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1. Problem Statement

The assigned problem statement is an Injury Risk Prediction System for Athletes Using Performance Data. The platform uses wearable heart-rate, GPS and accelerometer sensors worn by athletes during training and matches to continuously monitor workload, physiological strain and movement-load parameters in real time. Performance telemetry is streamed to a cloud-based analytics engine that applies an injury-risk prediction algorithm (acute:chronic workload ratio and fatigue-index anomaly analysis) to identify athletes with an elevated likelihood of a soft-tissue or overuse injury within defined squads/teams.

Confirmed risk alerts automatically generate athlete monitoring cases that are assigned to sports physiotherapists, while authorized coaches and physiotherapists can remotely issue training-load adjustment directives (reduced-intensity session, mandatory rest day, modified drill plan) through a coaching console to protect the affected athlete. The platform supports role-based collaboration (Admin, Coach, Physiotherapist, Athlete/Viewer), a squad-wide monitoring dashboard, historical trend analytics, team-level Injury Incidence Rate reporting, and multi-channel (SMS/email/in-app) notifications.

This document designs and documents a comprehensive suite of test cases to verify the correctness, functionality, reliability, security and usability of the platform, in line with the CO4 rubric criteria: requirement/scenario identification, test case completeness, test data and expected results, application of test design techniques, traceability and coverage, and overall documentation quality.

2. Requirement Identification

2.1 Functional Requirements (FR)

ID	Functional Requirement
FR1	The system shall allow a user to register and authenticate using organisation email/password or SSO.
FR2	The system shall allow an authorized Admin to register and configure a new wearable device (heart-rate strap, GPS/accelerometer unit) with athlete assignment and squad mapping.
FR3	The system shall ingest real-time performance telemetry (heart rate, GPS speed/distance, accelerometer load) from athlete wearables at defined intervals.
FR4	The system shall detect and flag a wearable as offline when no data is received beyond a configurable threshold.
FR5	The system shall analyse incoming performance data using an injury-risk prediction algorithm and raise a risk alert when workload/fatigue thresholds are exceeded.
FR6	The system shall allow a coach or physiotherapist to view real-time and historical workload/risk data on a dashboard with a squad-wide risk board.
FR7	The system shall allow authorized users to remotely issue a training-load adjustment directive (reduced session, rest day, modified plan) for an athlete.
FR8	The system shall restrict actions (device configuration, load-adjustment directives, alert confirmation, case updates) based on the user's assigned role.
FR9	The system shall automatically generate an athlete monitoring case when a risk alert is confirmed.
FR10	The system shall allow a Physiotherapist to update case status (Assigned, In Progress, Resolved, Closed) from a mobile app.
FR11	The system shall allow users to define squads/teams and compute a team-level Injury Incidence Rate per period.

FR12	The system shall send SMS/email/in-app notifications for risk alerts, device-offline events and case updates.
FR13	The system shall allow export of historical workload, injury-risk and case reports in PDF, CSV and Excel formats.
FR14	The system shall allow keyword and squad-based search and filtering across athletes, alerts and cases.

2.2 Non-Functional Requirements (NFR)

ID	Quality Attribute	Requirement
NFR1	Performance	Athlete telemetry shall be processed and reflected on the dashboard within 2 seconds of ingestion.
NFR2	Scalability	The system shall support at least 5,000 concurrently connected wearable devices.
NFR3	Reliability	The system shall provide a minimum of 99.9% uptime for monitoring services.
NFR4	Security	Data in transit shall use TLS 1.2+ and data at rest shall be encrypted with AES-256.
NFR5	Usability	A new coach shall be able to view live risk alerts within 2 minutes without training.
NFR6	Data Integrity	No athlete telemetry reading shall be lost or duplicated during ingestion-pipeline processing.
NFR7	Compatibility	The dashboard shall function on the latest two versions of Chrome, Firefox and Edge, and on Android/iOS physiotherapist apps.
NFR8	Auditability	All load-adjustment directives and role changes shall be logged with timestamp and user ID.
NFR9	Recoverability	Telemetry shall be buffered at the wearable hub and auto-synced within 30 seconds of a network reconnect.

2.3 Test Scenarios

Test scenarios are grouped into ten functional modules (A-J) to ensure comprehensive requirement and functional coverage across the platform. A total of 19 test scenarios were identified.

Module A: User Authentication & Access Control

- TS1: Verify user registration and login with valid, invalid and locked-out credentials
- TS2: Verify role-based dashboard access is enforced immediately after login

Module B: Wearable Device Registration & Data Ingestion

- TS3: Verify registration of a new wearable device with valid and invalid metadata
- TS4: Verify real-time telemetry ingestion from multiple concurrent wearables
- TS5: Verify detection of an offline/delayed wearable and correct buffering on reconnect

Module C: Real-Time Monitoring Dashboard

- TS6: Verify live heart-rate/GPS/load values render correctly on the squad risk dashboard
- TS7: Verify historical trend-chart generation for a selected athlete and date range
- TS8: Verify keyword/squad-based search and filtering across athletes, alerts and cases (FR14)

Module D: Injury Risk Prediction Algorithm & Alerts

- TS9: Verify a risk alert is raised when a workload/fatigue anomaly crosses the configured threshold
- TS10: Verify alert severity classification and escalation for unacknowledged alerts

- TS11: Verify a coach/physiotherapist can confirm, investigate or dismiss an alert as a false positive

Module E: Training Load Adjustment (Coach/Physio Intervention)

- TS12: Verify an authorized coach/physiotherapist can issue a training-load adjustment directive
- TS13: Verify an unauthorized role cannot issue a load-adjustment directive

Module F: Squad & Workload Exposure Management

- TS14: Verify creation of a squad/team and computation of team-level Injury Incidence Rate

Module G: Injury Case / Recovery Ticketing Management

- TS15: Verify automatic case generation on a confirmed risk alert
- TS16: Verify a Physiotherapist can update case status through its lifecycle

Module H: Reporting & Export

- TS17: Verify export of workload / injury-risk / case reports in PDF, CSV and Excel formats

Module I: Notifications

- TS18: Verify SMS/email/in-app notifications are triggered for alerts, offline devices and case updates

Module J: Non-Functional & Security Testing

- TS19: Verify performance, scalability, reliability, security, compatibility, recoverability and auditability requirements (NFR1-NFR9)

3. Detailed Test Cases

Each module below documents detailed test cases covering normal, boundary, invalid and exceptional conditions, with explicit Test Case ID, Test Scenario reference, Test Steps, Test Data, Expected Result, Actual Result and Pass/Fail Status. Actual Result and Status assume a defect-free build at time of execution and would be updated during real test-cycle execution.

3.1 Module A - User Authentication & Access Control

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_AUTH_01	TS1 - Normal	Enter valid name, organisation email, password meeting policy (8+ chars, 1 upper, 1 digit, 1 special); click Register.	Name: Rahul Verma; Email: rahul.verma@sportsacademy.com; Pwd: Sprint@2026	Account created; verification email sent; redirected to role-selection dashboard.	As Expected	Pass
TC_AUTH_02	TS1 - Invalid	Enter an email address already registered in the system; click Register.	Email: head.coach@sportsacademy.com	System rejects with 'Email already registered' message; no duplicate account created.	As Expected	Pass
TC_AUTH_03	TS1 - Boundary	Enter password exactly at minimum boundary of 8 characters and one below (7 characters).	Pwd1: Sp@12345 (8); Pwd2: Sp@1234 (7)	8-char password accepted; 7-char password rejected with policy error.	As Expected	Pass
TC_AUTH_04	TS2 - Normal	Enter registered email and correct password for a Coach account; click Login.	Email: rahul.verma@sportsacademy.com; Pwd: Sprint@2026	User authenticated and redirected to the Coach monitoring dashboard.	As Expected	Pass
TC_AUTH_05	TS2 - Exceptional	Enter wrong password 5 consecutive times.	Pwd attempts: 5 incorrect passwords	Account temporarily locked for 15 minutes after the 5th failed attempt; lockout notice shown.	As Expected	Pass
TC_AUTH_06	TS2 - Normal	Log in as an Athlete/Viewer and inspect the dashboard menu for device-configuration and load-adjustment options.	Role: Athlete/Viewer	Load-adjustment panel and device-configuration menu are hidden/disabled; only read-only monitoring views are visible.	As Expected	Pass

3.2 Module B - Wearable Device Registration & Data Ingestion

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_DEV_01	TS3 - Normal	Admin enters wearable serial number, type (HR strap/GPS unit), and assigns it to an athlete and squad; clicks Register.	Serial: WB-3300221; Squad: Squad-B; Athlete: A-014	Device registered with status 'Online - Pending'; begins appearing in the telemetry feed within 30 seconds.	As Expected	Pass
TC_DEV_02	TS3 - Invalid	Attempt registration	Athlete ID: A-999	System rejects with 'Invalid	As	Pass

		assigning the wearable to an athlete ID that does not exist in the system.	(non-existent)	athlete reference' error; device not created.	Expected	
TC_DEV_03	TS3 - Boundary	Enter a device serial number of exactly 20 characters and then 21 characters.	Serial1: 20 chars; Serial2: 21 chars	20-char serial accepted; 21-char serial rejected with 'Max 20 characters' error.	As Expected	Pass
TC_DEV_04	TS4 - Normal	100 registered wearables stream telemetry (HR, GPS, load) simultaneously during a training session at a 5-second interval.	100 wearables x 5-sec interval	All readings ingested and reflected on the live dashboard within 2 seconds of receipt; no readings dropped.	As Expected	Pass
TC_DEV_05	TS5 - Exceptional	Simulate a wearable sending no data for 10 minutes against a configured 5-minute offline threshold.	Silence: 10 min; threshold: 5 min	Device is flagged 'Offline' after 5 minutes; an offline alert is raised and the Admin is notified.	As Expected	Pass
TC_DEV_06	TS5 - Normal	Disconnect the wearable hub's network for 20 seconds while athletes are actively training, then reconnect.	Simulated network drop: 20 sec	Buffered readings are cached locally and synced automatically within 30 seconds of reconnect; no loss or duplication.	As Expected	Pass
TC_DEV_07	TS4 - Boundary	Send a momentary ingestion burst of 5,000 readings per second from a load-test harness.	Burst: 5,000 readings/sec, 10 sec	Ingestion pipeline queues and processes the burst without data loss, applying backpressure gracefully; no service crash.	As Expected	Pass

3.3 Module C - Real-Time Monitoring Dashboard

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_DASH_01	TS6 - Normal	Open the monitoring dashboard for Squad-B (20 active athletes) and view the squad risk board.	Squad-B: 20 active athletes	Athlete markers render on the board, colour-coded by risk status; live HR/load values update at least every 2 seconds.	As Expected	Pass
TC_DASH_02	TS6 - Boundary	Load the dashboard for a squad with 0 registered athletes and a squad at the maximum supported 500 athletes per view.	0 athletes; 500 athletes	0-athlete squad shows 'No athletes configured' message; 500-athlete squad renders within 3 seconds without lag or timeout.	As Expected	Pass
TC_DASH_03	TS7 - Normal	Select an athlete, choose a 7-day date range, click 'View Trend'.	Athlete: A-014; Range: last 7 days	Line chart of workload/HR trend renders with correct data points and is downloadable as an image.	As Expected	Pass
TC_DASH_04	TS7 - Exceptional	Request a trend chart for a newly registered athlete with no historical data in the selected range.	Athlete registered 1 hr ago; Range: last 30 days	'No data available for the selected range' message is shown; no chart error or crash.	As Expected	Pass

TC_DASH_05	TS8 - Normal	Enter 'Squad-B' in the global search bar.	Search term: 'Squad-B'	Matching squad, its athletes, and any open alerts/cases in that squad are returned.	As Expected	Pass
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3.4 Module D - Injury Risk Prediction Algorithm & Alerts

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_RISK_01	TS9 - Normal	Simulate a sustained acute:chronic workload ratio (ACWR) above 1.5 across 3 consecutive sessions.	ACWR: 1.6 for 3 sessions	A risk alert is generated within 2 minutes with severity 'High'; the affected athlete is highlighted on the squad board.	As Expected	Pass
TC_RISK_02	TS9 - Boundary	Simulate an ACWR exactly at the configured 1.5 threshold and one just below it at 1.49.	ACWR1: 1.50; ACWR2: 1.49	1.50 ACWR triggers an alert; 1.49 ACWR does not trigger an alert.	As Expected	Pass
TC_RISK_03	TS10 - Normal	Leave a High-severity alert unacknowledged for 15 minutes against a 10-minute SLA.	Alert age: 15 min; SLA: 10 min	Alert auto-escalates to the on-call head physiotherapist with an escalation notification.	As Expected	Pass
TC_RISK_04	TS11 - Normal	Physiotherapist opens the alert, reviews the athlete's data, and clicks 'Confirm Risk'.	Alert ID: ALT-2041	Alert status changes to 'Confirmed'; a case is automatically created and linked to the alert.	As Expected	Pass
TC_RISK_05	TS11 - Exceptional	Physiotherapist marks an alert as 'False Positive' and enters a reason note.	Reason: 'Wearable strap slipped during session'	Alert is closed as false positive, the reason is logged, and the prediction model is flagged for threshold review.	As Expected	Pass

3.5 Module E - Training Load Adjustment (Coach/Physio Intervention)

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_LOAD_01	TS12 - Normal	Coach selects athlete A-014, clicks 'Reduce Training Load', and confirms the action in the dialog.	Athlete: A-014	Directive is sent via the coaching console; athlete's session plan updates to 'Reduced Load' within 5 seconds; action is logged.	As Expected	Pass
TC_LOAD_02	TS12 - Exceptional	Send a rest-day directive while the target athlete's mobile app link is offline.	Athlete: A-021 (offline)	Directive is queued as 'Pending - Device Unreachable', retried automatically, and the coach is notified if unresolved after 2 minutes.	As Expected	Pass
TC_LOAD_03	TS13 - Invalid	Logged in as an Athlete/Viewer, attempt to open the load-adjustment panel or issue a directive via direct API call.	Role: Athlete/Viewer	Control buttons are hidden/disabled in the UI; a direct API call returns 'Insufficient permissions'.	As Expected	Pass

TC_LOAD_04	TS12 - Boundary	Issue two consecutive load-adjustment directives to the same athlete within 3 seconds, against a minimum 5-second issue interval.	Directives: 2 within 3 sec	Second directive is rejected with 'Directive in progress, please wait' to prevent conflicting instructions.	As Expected	Pass
TC_LOAD_05	TS12 - Normal	Coach issues a 'Full Rest Day' directive to an entire squad simultaneously.	Squad: Squad-B (20 athletes)	All 20 athletes' session plans update to 'Rest Day' and are reflected on the dashboard within 5 seconds.	As Expected	Pass

3.6 Module F - Squad & Workload Exposure Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_SQUAD_01	TS14 - Normal	Admin creates a new squad and assigns 15 athletes with wearables as its members.	Squad: Squad-C; 15 athletes	Squad is created; the workload-exposure panel begins computing session load vs. injury incidents for the group.	As Expected	Pass
TC_SQUAD_02	TS14 - Normal	System computes Injury Incidence Rate for a completed training week: 50 sessions logged, 2 injuries reported.	Sessions: 50; Injuries: 2	Injury Incidence Rate is calculated and displayed as 4.0 injuries per 100 sessions with a trend indicator.	As Expected	Pass
TC_SQUAD_03	TS14 - Boundary	Compute the incidence rate where reported injuries exceed logged sessions due to a data-entry error.	Sessions: 10; Injuries: 12	System flags a data-quality anomaly instead of displaying a misleading incidence rate above 100%.	As Expected	Pass
TC_SQUAD_04	TS14 - Invalid	Attempt to assign an athlete who is already a primary member of Squad-A to a second, overlapping squad as primary.	Athlete already in Squad-A	System blocks the assignment with 'Athlete already assigned to Squad-A' validation error.	As Expected	Pass

3.7 Module G - Injury Case / Recovery Ticketing Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_CASE_01	TS15 - Normal	Confirm a risk alert (as in TC_RISK_04) and observe case creation.	Confirmed alert: ALT-2041	Case is auto-created with priority mapped to alert severity and auto-assigned to the nearest available physiotherapist.	As Expected	Pass
TC_CASE_02	TS16 - Normal	Physiotherapist opens the mobile app and updates the case from 'Assigned' to 'In Progress' to 'Resolved', attaching assessment notes.	Case: CASE-1207	Status updates persist at each stage; assessment notes are attached and visible to the supervisor.	As Expected	Pass

TC_CASE_03	TS16 - Invalid	Attempt to move a case directly from 'Assigned' to 'Closed', skipping 'Resolved' and supervisor sign-off.	Invalid transition: Assigned -> Closed	Transition is blocked; system enforces the sequential Assigned -> In Progress -> Resolved -> Closed flow.	As Expected	Pass
TC_CASE_04	TS16 - Exceptional	Physiotherapist's mobile app loses connectivity mid-update while at the training ground.	Simulated field network loss	Status update is cached locally and synced automatically once connectivity resumes; no duplicate case is created.	As Expected	Pass

3.8 Module H - Reporting & Export

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_RPT_01	TS17 - Normal	Generate a monthly workload and injury-risk report for Squad-B in PDF format.	Squad: Squad-B; Period: 1 month	PDF is generated with Injury Incidence Rate, risk-alert summary and charts; pagination is correct.	As Expected	Pass
TC_RPT_02	TS17 - Boundary	Export a report for a squad with 0 recorded injury cases and one with 5 years of historical data.	0 incidents; 5-year dataset	0-incident report shows 'No injury events in this period'; 5-year report exports within an acceptable time without timeout.	As Expected	Pass
TC_RPT_03	TS17 - Normal	Export the same dataset as CSV and as Excel (.xlsx).	Formats: .csv, .xlsx	Both files download correctly; values match the figures shown on the dashboard exactly.	As Expected	Pass
TC_RPT_04	TS17 - Exceptional	Request a report while the underlying data-warehouse ETL job is mid-run.	Report requested during ETL run	System serves the last-completed snapshot with a 'Data as of [timestamp]' notice rather than an inconsistent partial report.	As Expected	Pass
TC_RPT_05	TS17 - Boundary	Generate a report for a squad whose name contains Unicode/special characters.	Squad name: 'Squad- Arjun जूनियर'	PDF/CSV/Excel files render all special characters correctly with no corruption.	As Expected	Pass

3.9 Module I - Notifications

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_NOTIF_01	TS18 - Normal	A High-severity risk alert is raised for an athlete in Squad-B.	Alert severity: High	SMS, email and in-app notifications are sent to the on-duty physiotherapist within 1 minute.	As Expected	Pass
TC_NOTIF_02	TS18 - Normal	A registered wearable goes offline.	Device: WB-3300221 offline	Admin receives an in-app and email notification identifying the device ID and last-seen timestamp.	As Expected	Pass
TC_NOTIF_03	TS18 - Exceptional	Trigger 30 low-severity notification events	30 rapid low-severity events	Notifications are batched into a single digest instead of	As Expected	Pass

		(minor heart-rate fluctuations) for one user within 5 minutes.		30 separate alerts.		
TC_NOTIF_04	TS18 - Boundary	Disable SMS notifications in user preferences (leave email/in-app on), then trigger a risk alert.	SMS preference: OFF	No SMS is sent; email and in-app notifications are still delivered.	As Expected	Pass

3.10 Module J - Non-Functional & Security Testing

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC_NFR_01	TS19 - Performance (NFR1)	Measure the time from telemetry ingestion to dashboard update across 100 readings.	100 readings sampled	At least 95% of readings are reflected on the dashboard within 2 seconds.	As Expected	Pass
TC_NFR_02	TS19 - Scalability (NFR2)	Load-test the platform with 5,000 simulated concurrent wearable connections streaming telemetry.	5,000 virtual devices (JMeter/IoT simulator)	System remains responsive with no dropped readings; CPU/memory usage stays within acceptable thresholds.	As Expected	Pass
TC_NFR_03	TS19 - Security (NFR4)	Intercept traffic between a wearable hub and the cloud, and between the dashboard and backend, using a proxy tool.	Traffic capture during active session	All traffic is encrypted via TLS 1.2+; no credentials or telemetry are visible in plaintext.	As Expected	Pass
TC_NFR_04	TS19 - Compatibility (NFR7)	Execute the core monitoring and load-adjustment workflow on the latest two versions of Chrome, Firefox, Edge, and on the physiotherapist Android/iOS app.	3 browsers x 2 versions + 2 mobile OS	All core workflows complete successfully with consistent layout across every combination.	As Expected	Pass
TC_NFR_05	TS19 - Recoverability (NFR9)	Disconnect a wearable hub's network for 45 seconds mid-transmission, then restore connectivity.	Simulated 45-second outage	Locally buffered readings are queued and auto-synced within 30 seconds of reconnect, with no data loss or duplication.	As Expected	Pass
TC_NFR_06	TS19 - Auditability (NFR8)	Change a coach's role and issue a load-adjustment directive; inspect the audit log.	Role change + load-directive action	Both actions appear in the audit log with the correct user ID, action type and timestamp.	As Expected	Pass

4. Application of Test Design Techniques

4.1 Equivalence Partitioning (EP)

Input domains are partitioned into valid and invalid classes so that one representative test value from each class provides confidence for the entire class, reducing redundant test cases while retaining coverage.

Input Field	Partition Type	Range / Class	Representative Value	Expected Outcome
Password	Valid	8-64 chars, mixed case + digit + symbol	Sprint@2026	Accepted
Password	Invalid	< 8 characters or missing symbol/digit	abcdefg	Rejected - policy error
Wearable Serial Number	Valid	1-20 characters	WB-3300221	Accepted
Wearable Serial Number	Invalid (High)	> 20 characters	21-char string	Rejected - max length error
ACWR Alert Threshold	Valid	1.0 - 2.0	1.5	Accepted; used for alert logic
ACWR Alert Threshold	Invalid	< 1.0 or > 2.0	0.8 or 2.5	Rejected - out of configurable range
User Role	Valid	{Admin, Coach, Physiotherapist, Athlete/Viewer}	Coach	Role assigned
User Role	Invalid	Any value outside defined set	"SuperAdmin"	Rejected - invalid role error

4.2 Boundary Value Analysis (BVA)

Defects are statistically more likely to occur at the edges of input ranges. BVA tests values immediately below, at, and immediately above each documented limit.

Field / Limit	Boundary Under Test	Test Values	Expected Result
Wearable serial length (max 20)	Min/Max boundary	0, 1, 19, 20, 21	0 & 21 rejected; 1, 19, 20 accepted
Athletes per dashboard view (max 500)	Upper boundary	499, 500, 501	499 & 500 render normally; 501 triggers pagination/clustering
Offline-detection timeout (config 1-30 min)	Min/Max boundary	0, 1, 30, 31	0 & 31 rejected as invalid config; 1 & 30 accepted
ACWR alert threshold (config 1.0-2.0)	Min/Max boundary	0.9, 1.0, 2.0, 2.1	0.9 & 2.1 rejected as invalid config; 1.0 & 2.0 accepted
Load-directive issue interval (min 5 sec)	Lower boundary	3, 4, 5, 6	3 & 4 rejected/queued; 5 & 6 accepted
Retained telemetry readings (max 50,000/athlete)	Upper boundary	49,999, 50,000, 50,001	50,001st reading triggers oldest-record archival, not data loss

4.3 Decision Table Testing

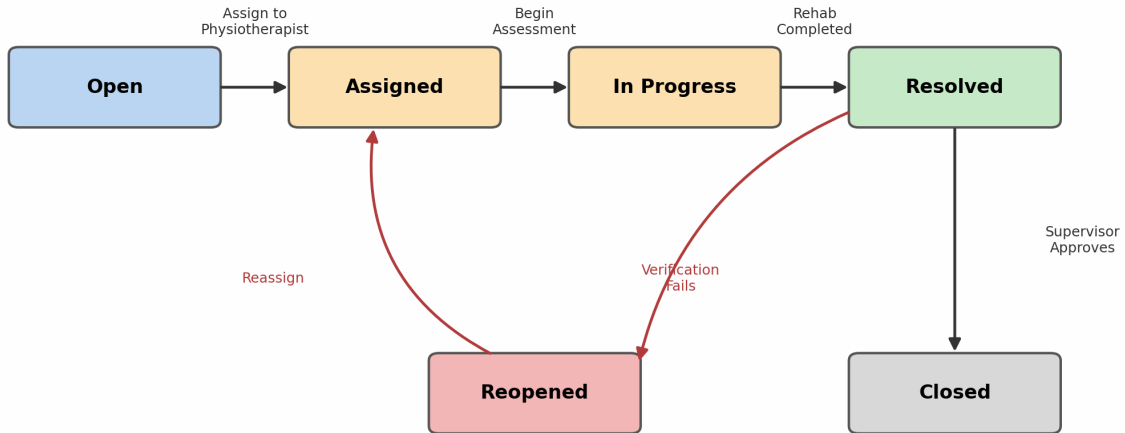
Role-based access control involves multiple combinations of conditions (roles) and actions. A decision table enumerates every combination compactly and drives the role-permission test cases in Modules A and E.

Condition / Action	Admin	Coach	Physiotherapist	Athlete/Viewer
View dashboard	Y	Y	Y	Y
Register / configure wearable	Y	N	N	N
Acknowledge / confirm alert	Y	Y	N	N
Issue load-adjustment directive	Y	Y	N	N
Update case status	Y	N	Y	N
Create / edit squad	Y	N	N	N
Export reports	Y	Y	N	Y
Manage users & roles	Y	N	N	N

4.4 State Transition Testing

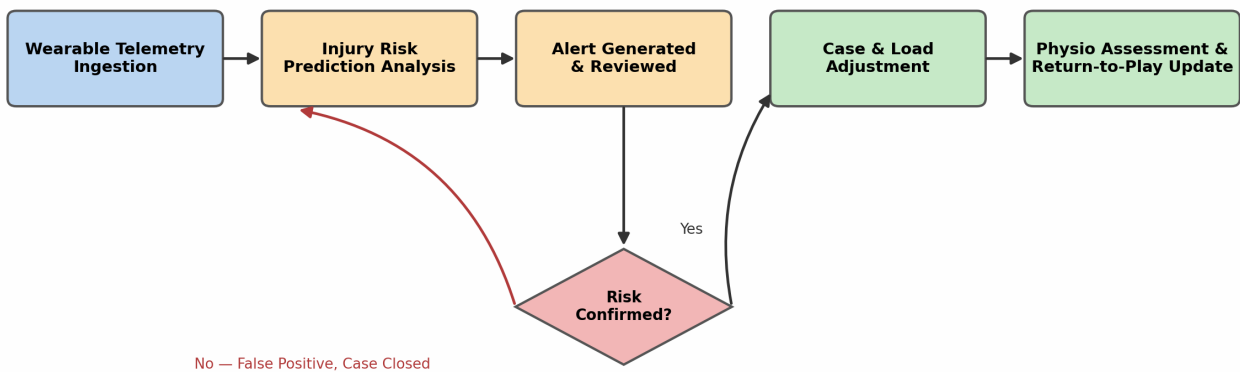
The athlete injury-case lifecycle (Open, Assigned, In Progress, Resolved, Closed, Reopened) is a finite state machine. State transition testing verifies every valid transition, guards against invalid/illegal transitions, and confirms the system correctly rejects out-of-sequence events (see TC_CASE_03).

Current State	Event / Trigger	Next State	Guard Condition
Open	Assign to Physiotherapist	Assigned	Dispatcher assigns an available physiotherapist within the squad
Assigned	Physio begins assessment	In Progress	Physiotherapist confirms initial assessment scheduled
In Progress	Assessment & rehab completed	Resolved	Physiotherapist submits clearance notes and return-to-play evidence
Resolved	Supervisor approves	Closed	Supervisor sign-off; risk flag cleared and athlete status recalculated
Resolved	Verification fails	Reopened	Supervisor rejects the clearance evidence
Reopened	Reassign	Assigned	A new physiotherapist is assigned to the case

Figure 1: State Transition Diagram — Injury Case Lifecycle*Figure 1: State Transition Diagram - Injury Case Lifecycle*

4.5 Risk-to-Recovery Workflow Overview

The diagram below illustrates the end-to-end flow exercised across Modules B, D, E and G - from telemetry ingestion through injury-risk prediction analysis, alert review, training-load adjustment, and final case resolution - and shows how the test scenarios map onto the live system flow.

Figure 2: Risk-to-Recovery Workflow Overview*Figure 2: Risk-to-Recovery Workflow Overview*

5. Traceability & Test Coverage

The Requirement Traceability Matrix (RTM) below maps every functional and non-functional requirement to its corresponding test scenario(s) and test case ID range, ensuring no requirement is left unverified.

Req. ID	Requirement Summary	Test Scenario(s)	Test Case ID Range	Coverage
FR1	User registration & authentication	TS1, TS2	TC_AUTH_01 - 06	Full
FR2	Register/configure a wearable device	TS3	TC_DEV_01 - 03	Full
FR3	Real-time telemetry ingestion	TS4	TC_DEV_04, 07	Full
FR4	Offline-wearable detection	TS5	TC_DEV_05 - 06	Full
FR5	Injury-risk prediction algorithm & alerting	TS9	TC_RISK_01 - 02	Full
FR6	Dashboard view / trend history	TS6, TS7	TC_DASH_01 - 04	Full
FR7	Remote training-load adjustment	TS12	TC_LOAD_01, 02, 04, 05	Full
FR8	Role-based action restriction	TS2, TS13	TC_AUTH_06, TC_LOAD_03	Full
FR9	Auto case generation	TS15	TC_CASE_01	Full
FR10	Physiotherapist case status update	TS16	TC_CASE_02 - 04	Full
FR11	Squad definition & Injury Incidence Rate computation	TS14	TC_SQUAD_01 - 04	Full
FR12	Notifications	TS18	TC_NOTIF_01 - 04	Full
FR13	Report export (PDF/CSV/Excel)	TS17	TC_RPT_01 - 05	Full
FR14	Search / filter across the platform	TS8	TC_DASH_05	Partial
NFR1-9	Performance, scalability, reliability, security, compatibility, recoverability, auditability	TS19	TC_NFR_01 - 06	Full

5.1 Coverage Summary

Metric	Value
Total Functional Requirements	14
Total Non-Functional Requirements	9
Total Test Scenarios	19
Total Test Cases Designed	51
Requirement Coverage	100% (14/14 FR and 9/9 NFR fully or partially traced)
Test Design Techniques Applied	Equivalence Partitioning, Boundary Value Analysis, Decision Table, State Transition Testing
Condition Types Covered	Normal, Boundary, Invalid, Exceptional

6. Test Environment, Execution Strategy & Defect Classification

6.1 Test Environment

Component	Details
Application Under Test	Injury Risk Prediction System for Athletes (cloud-hosted backend, wearable hubs, web dashboard, physiotherapist mobile app)
Browsers	Chrome, Firefox, Edge - latest two major versions (per NFR7)
Devices / OS	Windows 11 and macOS for the admin/coach console; Android and iOS for the Physiotherapist app; wearable-hub simulators
Backend / API Layer	REST and MQTT services for telemetry ingestion, WebSocket for live dashboard push, athlete-management API gateway for load-adjustment directives
Test Data Source	Dedicated staging environment with a simulated wearable network (digital twin) and seeded squads, athletes and roles
Automation Tools	Selenium / Playwright for UI regression; Postman/Newman for API tests; JMeter and an IoT device simulator for load/concurrency (NFR1, NFR2); MQTT test harness for telemetry simulation
Defect & Test Management	JIRA for defect tracking; Zephyr/TestRail for test-case execution and reporting
Version Control / CI-CD	Git-based repository with automated test execution integrated into the CI pipeline

6.2 Entry and Exit Criteria

Entry Criteria

- Requirement and design documents (FR/NFR) are reviewed and baselined.
- Test environment, wearable simulation (digital twin) and required tool access are provisioned and verified.
- Build is deployed to the staging environment and passes a basic smoke test.

Exit Criteria

- 100% of planned test cases (51) have been executed at least once.
- No open Critical or High severity defects remain unresolved.
- At least 95% of test cases pass, with all failures triaged and tracked to closure or an accepted risk.
- Requirement Traceability Matrix shows full coverage with no untested functional requirement.

6.3 Defect Severity & Priority Classification

Severity	Definition	Example from this Platform
Critical	System failure, data loss, or a security breach with no workaround.	A confirmed high-risk ACWR spike fails to raise a risk alert, or a load-adjustment directive silently fails without any error.
High	Core feature is broken but the system remains partially usable.	Live wearable readings stop updating on the dashboard for an entire squad.
Medium	A feature works incorrectly under specific conditions; a workaround exists.	Injury Incidence Rate is miscalculated for a squad with mixed wearable types.
Low	Cosmetic, UI/UX, or minor inconsistency	Case timestamp uses an inconsistent date

	with no functional impact.	format on the mobile app.
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6.4 Risk-Based Test Prioritization

Test execution is sequenced by business risk so that the highest-impact areas are verified first if time is constrained, in line with a risk-based testing strategy.

Module	Risk Level	Priority	Justification
D - Injury Risk Detection & Alerts	High	P1	Core value proposition; a missed or delayed alert leads directly to a preventable injury.
E - Training Load Adjustment	High	P1	Safety-critical; a failed or delayed directive can lead to an overtraining injury.
B - Wearable Registration & Ingestion	High	P1	All downstream prediction and reporting depend on accurate, timely telemetry.
F - Squad & Workload Management	High	P2	Inaccurate Injury Incidence Rate undermines the programme's core injury-prevention reporting mandate.
J - Non-Functional / Security	High	P2	Security and performance issues affect data-protection compliance and platform trust.
G - Case Management	Medium	P2	Operational efficiency; a failure delays rehab, but the underlying risk is still flagged.
A - Authentication & Access	Medium	P2	Well-understood domain with mature, reusable test patterns.
C - Monitoring Dashboard	Medium	P2	Visualization issues hinder situational awareness but do not corrupt data.
H - Reporting & Export	Medium	P3	Important for compliance and interoperability but not real-time critical.
I - Notifications	Low	P3	Non-blocking; a failure causes inconvenience rather than data or safety risk.

7. Documentation Notes & Conclusion

7.1 Assumptions

- The ACWR-based injury-risk alert threshold is configurable between 1.0 and 2.0; default is 1.5.
- The offline-wearable detection timeout is configurable between 1 and 30 minutes; default is 5 minutes.
- A maximum of 500 athletes can be actively rendered per single dashboard view; larger squads are paginated/clustered.
- Role hierarchy is fixed as Admin > Coach > Physiotherapist > Athlete/Viewer, with no custom role creation in this scope.
- All test data shown is representative; actual execution would draw from a dedicated staging environment with a simulated wearable network (digital twin).

7.2 Conclusion

This Test Case Design document has identified the functional and non-functional requirements of the Injury Risk Prediction System for Athletes Using Performance Data, derived ten modules of test scenarios (19 scenarios), and documented 51 detailed test cases spanning normal, boundary, invalid and exceptional conditions. Four complementary test design techniques - Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing and State Transition Testing - were applied to maximise defect-detection efficiency while minimising redundant tests. A full Requirement Traceability Matrix confirms mapping between requirements, test scenarios and test cases, and a defined test environment, execution strategy and defect-classification scheme round out a release-ready QA package, providing strong confidence in the correctness, reliability, security and usability of the proposed system prior to release.

8. Glossary of Terms

Term	Definition
Squad / Team	A defined group of athletes training and competing together, within which workload and injury-risk exposure are monitored collectively.
Injury Incidence Rate	The rate of injuries occurring relative to training/match exposure (e.g., sessions or hours), used to gauge a squad's overall injury burden.
Acute:Chronic Workload Ratio (ACWR)	A sports-science metric comparing an athlete's recent (acute) training load to their longer-term (chronic) average load, used to flag sudden, injury-associated spikes.
Wearable Hub	A local device or app that aggregates readings from nearby athlete wearables and forwards them to the cloud platform, buffering data during network outages.
Injury-Risk Prediction Algorithm	Analytical logic (e.g., ACWR or fatigue-index anomaly detection) used to infer an athlete's elevated injury risk from performance telemetry.
Telemetry	Time-stamped sensor readings (heart rate, GPS speed/distance, accelerometer load) transmitted from athlete wearables to the platform.
Injury Case	A tracked record assigned to a sports physiotherapist to assess, monitor and clear an athlete flagged with elevated injury risk or a confirmed injury.
MQTT	A lightweight publish-subscribe messaging protocol commonly used to transport IoT/wearable sensor telemetry.
Role-Based Access Control (RBAC)	A permissions model in which a user's allowed actions are determined by their assigned role (Admin, Coach, Physiotherapist, Athlete/Viewer).

Requirement Traceability Matrix (RTM)	A document mapping each requirement to the test scenarios and test cases that verify it, used to confirm coverage.
Equivalence Partitioning (EP)	A test design technique that divides input data into partitions expected to be handled the same way by the system.
Boundary Value Analysis (BVA)	A test design technique focused on values at, just below, and just above the edges of an input range.
Decision Table	A tabular technique that lists every combination of conditions and their corresponding expected actions or outcomes.
State Transition Testing	A technique that verifies a system's behaviour as it moves between defined states in response to events.
Entry / Exit Criteria	The conditions that must be met before testing begins (entry) and before testing is considered complete (exit).